Vincent Zhen

(416 -897-0955) | vhzhen@uwaterloo.ca | linkedin.com/in/vincent-zhen-190885351 | github.com/vhzhen | vhzhen.vercel.app

Education

University of Waterloo GPA: 3.98/4

Computer Science and Finance Double Major (BCFM) *Expected: April 2029*

- KPMG Scholarship ($2000) and the University of Waterloo President's Scholarship of Distinction ($2000)
- Junior Analyst in the University of Waterloo Student Venture Fund (SVF)
- Teaching Assistant for CFM 101, collaborating with the professor on new course materials

Experience

Full-Stack Software Engineer Kitchener, ON

Carta *Present*

- Implemented a backend service and data pipeline using **Kafka**, **Python**, **Django** and **PostgreSQL**, computing over **60,000** cap tables daily into materialized views to solve an **N+1 query problem** and power Carta's **MCP**
- Rebuilt the share accounting engine in **Django** with **gRPC**, **Kubernetes** and **Redis** for **20M** certificates and **31M** vesting events, cutting the slowest **5%** of cap table loads **from 250 ms to 40 ms** across **5M** weekly loads
- Piloted a **React** and **Redux** compliance dashboard tracking Rule 701 grant limits and QSBS holding periods across **500** companies, flagging at-risk issuances and cutting manual spreadsheet checks
- Initiated 1:1s with engineers across the Kitchener-Waterloo, San Francisco and Rio offices to identify outstanding tasks and pain points. Led the effort to remove stale feature flags, simplifying conditional logic across **Python**, **Django**, and **TypeScript** to reduce code complexity

ETF Trader Toronto, ON

TD Securities *January 2026 – April 2026*

- Built and iterated a **Python** analytics tool (**PyQt5**, **pyodbc**, **Microsoft SQL**, **xbbg**) to analyze spread inefficiencies across orders and adjust quoting strategies, incorporating feedback from traders into the GUI design
- Engineered a Financial Information eXchange protocol app with **Python**, **PyQt5** and **Solace Message Queue** to enable direct electronic communication for the creation and redemption of ETFs
- Refactored a **Visual Basic** ETF parser into a **Python** and **Jenkins** workflow using **Claude**, allowing close price checking via **Bloomberg API** while integrating **SQL** database access and logging

Software Developer Toronto, ON

Pepper *April 2025 – August 2025*

- Collaborated with four developers and customers to build an AI-powered website creation pipeline using **Python**, **n8n**, **Snowflake**, and **10Web**, securing **$10,000+** ARR in distributor contracts and advancing **10+** leads
- Spearheaded a large-scale image reordering project using **OpenAI Vision** to reorganize **1M+ images** across **400,000+** unique food products, improving thumbnail image accuracy to over **95%**
- Designed and deployed ETL pipelines, ingesting **450,000+** items data (images, allergens, etc) from vendors like Syndigo, Frosty Acres, and Essendant via **API/SFTP** into **MongoDB** and **AWS S3**, with scheduled **AWS Glue** runtimes and **CI/CD** deployment through **GitHub Actions**

Projects

WatBal | *Flutter, Dart, Kotlin, Swift* | GitHub

- Built a cross-platform Android/iOS app that displays University of Waterloo balances, account data and transaction history in-app and via home-screen widgets, with skeleton loading screens and micro-animations
- Added background refresh via WorkManager and BGAppRefreshTask so widgets update without opening the app

notifco | *Go, PostgreSQL, Redis, Docker, Prometheus* | GitHub

- Solo-developed a Go Discord bot serving **20+ servers** that DMs University of Waterloo students within **60 seconds** of a full course section opening, replacing UW Flow's tool that had up to **28 hours** of delay
- Designed idempotent alert delivery using Redis compare-and-set for atomic edge detection and a Postgres outbox with SKIP LOCKED workers, preventing duplicate DMs via a unique key on subscription and transition
- Deployed with Docker and GitHub Actions CI, with Prometheus metrics, and open-sourced under MIT

Technical Skills

Languages: Python, TypeScript, SQL, C, C++, Visual Basic, HTML/CSS, Bash
Frameworks & Libraries: Django, React, Redux, Next.js, Flutter, PyQt5, pandas, NumPy, Matplotlib
Databases & Messaging: PostgreSQL, Microsoft SQL Server, MongoDB, Redis, Snowflake, Kafka
Cloud & Developer Tools: AWS, Kubernetes, gRPC, Jenkins, GitHub Actions, Git, Bloomberg API, OpenAI, n8n